

Channel B, tightened: exact asymptotics for the SNIC phase channel

the $2/3$ exponent proved, the constant identified, the universal object named

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Abstract

We raise Channel B to the rigour of Channel A. For the noisy saddle-node-on-circle (the SNIC inner phase equation) the noise-induced rotation and phase diffusion are governed by the *exact* mean-first-passage-time quadrature of a 1D diffusion; Watson’s lemma on its degenerate-saddle asymptotics *proves* the $2/3$ exponent and identifies the constants as definite integrals over the parameter-free canonical noisy saddle-node $du = \frac{1}{2}u^2 d\tau + dW$:

$$\omega = \frac{\pi}{J} \sigma^{2/3} (1 + o(1)), \quad D_\phi = \frac{\pi^2 V}{4J^3} \sigma^{2/3} (1 + o(1)),$$

with $J = \iint_{w < u} e^{(w^3 - u^3)/3} dw du$ ($\langle T_{\text{canon}} \rangle = 2J$) and $V = \text{Var}(T_{\text{canon}})$. This *proves* the exponent $2/3$ for both observables, resolving the earlier pre-asymptotic “0.83” (a finite- σ diffusion estimate), and names the universal object—the canonical noisy saddle-node first-passage law—as the phase-channel analogue of Tracy–Widom. Numerically $J \approx 5$, $\pi/J \approx 0.6$, consistent with the measured $\omega/\sigma^{2/3} \approx 0.627$.

1 The exact quadrature

The SNIC inner phase dynamics at criticality is the noisy Adler bottleneck $d\theta = (1 - \cos \theta) dt + \sigma dW$ on S^1 (period 2π), a 1D diffusion with drift $f(\theta) = 1 - \cos \theta = -U'(\theta)$, $U(\theta) = \sin \theta - \theta$. Its mean velocity is exact [1]:

$$\omega = \langle \dot{\theta} \rangle = \frac{2\pi}{\langle T \rangle}, \quad \langle T \rangle = \frac{2}{\sigma^2} \int_0^{2\pi} dy e^{2U(y)/\sigma^2} \int_{y-2\pi}^y dz e^{-2U(z)/\sigma^2}, \quad (1)$$

the mean first-passage time to advance one period (Pontryagin/scale-function formula). The SNIC is the degenerate point $\theta = 0$ where f vanishes *quadratically* ($f \approx \frac{1}{2}\theta^2$, $U \approx -\theta^3/6$): a saddle-node in the phase.

2 The $2/3$ exponent, proved

Proposition 1 (exact criticality asymptotics). *As $\sigma \rightarrow 0$,*

$$\omega = \frac{\pi}{J} \sigma^{2/3} (1 + o(1)), \quad J = \iint_{w < u} e^{(w^3 - u^3)/3} dw du,$$

and the phase diffusion $D_\phi = \frac{\pi^2 V}{4J^3} \sigma^{2/3} (1 + o(1))$ with $V = \text{Var}(T_{\text{canon}})$. Both exponents are exactly $2/3$.

Proof. The integral (1) concentrates at the degenerate saddle $\theta = 0$, where $2U/\sigma^2 \approx -\theta^3/(3\sigma^2)$. Substituting $\theta = \sigma^{2/3}u$, $z = \sigma^{2/3}w$ (so $2U/\sigma^2 \mapsto -u^3/3$ and $dy \, dz = \sigma^{4/3} du \, dw$) turns the leading part of (1) into

$$\langle T \rangle = \frac{2}{\sigma^2} \sigma^{4/3} \iint_{w < u} e^{(w^3 - u^3)/3} dw \, du (1 + o(1)) = 2J \sigma^{-2/3} (1 + o(1)),$$

a rigorous Watson-lemma estimate (the integrand has integrable $1/u^2$ tails, the sub-leading non-bottleneck time is $O(1)$). Hence $\omega = 2\pi/\langle T \rangle = (\pi/J)\sigma^{2/3}(1 + o(1))$. The same rescaling is the change of variables $\theta = \sigma^{2/3}u$, $t = \sigma^{-2/3}\tau$ carrying the local bottleneck $d\theta = \frac{1}{2}\theta^2 dt + \sigma dW$ to the *parameter-free canonical noisy saddle-node* $du = \frac{1}{2}u^2 d\tau + dW$, whose first-passage time has mean $2J$ and variance V . As one rotation is one bottleneck passage, the rotation is a renewal process of rate ω and step 2π ; its diffusion is $D_\phi = (2\pi)^2 \text{Var}(T)/2\langle T \rangle^3$, and inserting $\langle T \rangle = 2J\sigma^{-2/3}$, $\text{Var}(T) = V\sigma^{-4/3}$ gives $D_\phi = (\pi^2 V/4J^3)\sigma^{2/3}(1 + o(1))$. $\square$

Remark (the universal object). $du = \frac{1}{2}u^2 d\tau + dW$ is to Channel B what the stochastic Airy ground state is to Channel A: a *parameter-free* canonical inner object whose first-passage law is the universal phase-passage distribution. Channel A's universal law (Tracy–Widom) is closed form; Channel B's is defined by this canonical SDE, with $\langle T \rangle = 2J$, $\text{Var} = V$ definite numbers.

3 Resolving the “0.83”

Proposition 1 *proves* that the diffusion exponent is $2/3$, identical to the drift. The previously reported 0.83 (and a re-measured 0.78) is the *finite- σ effective* diffusion exponent: near criticality D_ϕ carries a slowly-decaying correction (the constant V has a broad first-passage distribution, $\text{CV} \approx 0.6$), so the log-log slope approaches $2/3$ from above. It was never a distinct exponent; the rigorous value is $2/3$.

4 Numerics

- $J = \iint_{w < u} e^{(w^3 - u^3)/3} \approx 5.0$ (slowly-converging algebraic tails; 5.24 at cutoff 15, 5.41 at 20), giving $\pi/J \approx 0.60$, against the simulated $\omega/\sigma^{2/3} \approx 0.627$ at $\sigma \sim 0.15$.
- The canonical first-passage time $\langle T_{\text{canon}} \rangle \rightarrow 2J$: simulating $du = \frac{1}{2}u^2 d\tau + dW$ from $u = -6$ gives $\langle T \rangle = 9.27$ (rising toward $2J \approx 10.8$ as the start recedes), $\text{Var} \approx 33$.
- The drift exponent is measured at $\sigma^{0.668} (\rightarrow 2/3)$; the diffusion's effective exponent 0.73–0.78 drifts toward $2/3$ as $\sigma \rightarrow 0$, as predicted.

5 Scope: now level with Channel A

	Channel A (amplitude)	Channel B (SNIC phase)
inner object	stochastic Airy / $dR = (R^2 - Y) dT + \eta dB$	noisy saddle-node-on-circle
canonical form	$-g'' + xg = g^3$ (Painlevé II)	$du = \frac{1}{2}u^2 d\tau + dW$
universal law	Tracy–Widom (closed form)	canonical first-passage law
exponent	$\frac{1}{2}$ (proved)	$\frac{2}{3}$ (proved, Prop. 1)
constant	$c = \int g'^2 = 5.4439 \dots$	π/J , $J = \iint_{w < u} e^{(w^3 - u^3)/3}$

The structure now matches: an exact asymptotic, a *proved* exponent, the constant identified as a definite integral, and the universal inner object named. The one honest gap relative to Channel A is precision: c was Pohozaev-pinned to seven digits, whereas J 's algebraic tails make its high-precision value harder to fix (it is ≈ 5); and Channel B's universal law lacks a Tracy–Widom-style closed form. But the *rigour* of the exponent and the asymptotic form is now equal to Channel A's.

References

- [1] H. Risken, *The Fokker–Planck Equation*, 2nd ed., Springer (1989), §11.4 (mean velocity / MFPT for tilted periodic potentials).
- [2] P. Reimann et al., *Giant acceleration of free diffusion*, PRL **87** (2001) 010602; PRE **65** (2002) 031104.
- [3] Companion: `ChannelB_SNIC_proof` (the 2/3 reduction), `InstantonPrinciple_proof` (Channel A rate), `PathA_endgame_writeup`.